# AIR POLLUTION CONTROL DETAILS

***(For MPCB – Consent to Establish / Consent to Operate)***

## 1. General Information

- Name of Unit: Akshay Sansaadhan Pvt. Ltd.
- Location: Godown No 21 & 22, Gat No 96/1, Charholi Khurd, Tal. Khed, Dist. Pune 412 105
- Proposed Activity: Processing of agricultural / organic waste into biomass pellets, briquettes, and liquid organic fertilizer (LOF)
- Plant Capacity: 5 TPD
- Number of Workers: 8 Nos.
- Consent Applied For: CTE / CTO

## 2. Sources of Air Emissions

The following are the potential sources of air emissions in the proposed unit:

| Sr. No. | Source | Nature of Emission |
|---|---|---|
| 1 | Shredder / Hammer mill | Dust |
| 2 | Screening & material transfer | Fugitive dust |
| 3 | Thermal dryer | Particulate matter |
| 4 | Biomass handling & storage | Fugitive dust |

## 3. Air Pollution Control Systems Provided

To control emissions, the unit shall install and operate the following Air Pollution Control Devices (APCDs):

| Source | APCD Provided |
|---|---|
| Hammer mill | Dust collector |
| Material transfer points | Enclosures & local suction |
| Thermal dryer exhaust | Cyclone separator |

| Source | APCD Provided |
|---|---|
| Post-cyclone exhaust | Water scrubber |
| Final emission stage | Bag filter |
| Overall system | Blower with adequate stack |

📌 **The APCDs are installed in series to ensure effective removal of particulate matter before discharge.**

**4. Description of Air Pollution Control System (Process-wise)**

Cyclone Separator

- Used as the primary dust removal device
- Removes coarse particulate matter from dryer exhaust gases

Water Scrubber

- Installed in Dryer Cum Mixture Unit
- Removes fine particulate matter
- Scrubbing water is recirculated in a closed loop

Bag Filter

- Installed as final polishing unit
- Ensures emissions are within permissible limits
- Collected dust is reused in the process

Dust Collector

- Provided at hammer mill and screening sections
- Prevents fugitive dust generation inside the shed

**5. Stack & Emission Control**

- Adequate height stack is provided after APCDs
- Emissions shall be maintained within limits prescribed by MPCB
- No black smoke or visible emissions shall be generated

**6. Fugitive Emission Control Measures**

- All processing equipment housed inside closed shed
- Regular water sprinkling for dust suppression
- Covered storage of raw material and finished goods
- Pucca flooring to avoid dust resuspension
- Proper housekeeping maintained at all times

**7. Scrubber Water & Leachate Management (Important)**

- Water scrubber operates in closed-loop recirculation mode
- No wastewater is discharged outside the premises
- Any collected condensate / leachate from organic waste:
    - Is collected separately
    - Stabilized and converted into Liquid Organic Fertilizer (LOF)
    - Utilized / sold as value-added product

📌 **Hence, no industrial effluent is generated, and the unit operates under Zero Liquid Discharge (ZLD) conditions.**

**8. Noise & Odour Control (linked to air management)**

- Acoustic enclosures for high-noise equipment
- Preventive maintenance of machinery
- No chemical additives used
- No odour nuisance anticipated

**9. Monitoring & Maintenance**

- APCDs shall be operated continuously during plant operation
- Periodic inspection and cleaning of cyclone, scrubber and bag filter
- Records of operation and maintenance shall be maintained

**10. Declaration**

**The unit shall provide and maintain adequate air pollution control systems.**
**Emissions shall be controlled as per MPCB norms.**
**The unit shall operate with minimum dust generation and Zero Liquid Discharge (ZLD).**

**For Akshay Sansaadhan Pvt. Ltd.**






Authorized Signatory
Name: Abhijeet Gengane
Designation: Founder & CEO
Date: 01-02-2026
Place: Pune

# NOISE CONTROL MEASURES

***(For MPCB – Consent to Establish / Consent to Operate)***

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## 2. Sources of Noise

The major sources of noise in the proposed unit are:

- Shredder / Hammer mill
- Pellet machine / Briquette machine
- Blowers and fans
- Screening equipment

**All machinery shall be installed inside an enclosed industrial shed.**

## 3. Noise Control Measures Provided

The following measures shall be implemented to control noise levels:

- Installation of machinery on anti-vibration pads / foundations
- Provision of acoustic enclosures for high-noise equipment, wherever required
- Regular preventive maintenance of machinery to avoid abnormal noise
- Proper lubrication of moving parts

**4. Operational Controls**

- Plant operation shall be restricted to daytime working hours
- No night-time operations shall be carried out unless permitted
- Machinery shall be operated only under rated capacity conditions

**5. Worker Safety Measures**

- Workers shall be provided with personal protective equipment (PPE) such as ear plugs / ear muffs, if required
- Workers shall be trained on safe operating practices

**6. Compliance with Noise Standards**

- Noise levels at plant boundary shall be maintained within limits prescribed under the Noise Pollution (Regulation and Control) Rules, 2000
- No noise nuisance shall be caused to nearby residential areas

**7. Monitoring & Maintenance**

- Periodic noise level checks shall be carried out
- Corrective actions shall be taken immediately if abnormal noise is observed

**8. Declaration**

**The unit shall implement adequate noise control measures. Noise levels shall be maintained within permissible limits as prescribed by MPCB and applicable rules.**

**For Akshay Sansaadhan Pvt. Ltd.**






Authorized Signatory
Name: Abhijeet Gengane
Designation: Founder & CEO
Date: 01-02-2026
Place: Pune

# ODOUR CONTROL MEASURES

***(For MPCB – Consent to Establish / Consent to Operate)***

## 1. General Information

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- Consent Applied For: CTE / CTO

## 2. Potential Sources of Odour

Odour may be generated from:

- Temporary storage of organic / wet biomass
- Shredding and dewatering of organic waste
- Handling of leachate / condensate prior to processing

## 3. Odour Control Measures Provided

The following measures shall be implemented to prevent odour nuisance:

- Prompt processing of incoming organic waste without long-term storage
- Covered storage of raw materials inside the shed
- Immediate collection of leachates and its conversion into Liquid Organic Fertilizer (LOF)
- Regular cleaning of processing areas
- No open dumping or accumulation of waste

## 4. Process & Design Controls

- The process is primarily dry and thermal, which inherently minimises odour generation
- No anaerobic digestion or putrefaction is involved
- No chemical additives or odour-causing substances are used

**5. Housekeeping & Management Practices**

- Daily removal of residues and rejects
- Maintenance of hygienic conditions within the premises
- Periodic inspection of waste handling areas

**6. Complaint Prevention & Redressal**

- In case of any odour complaint, immediate corrective action shall be taken
- The unit shall coordinate with local authorities, if required

**7. Declaration**

**The unit shall implement adequate odour control measures. No odour nuisance shall be caused to surrounding areas. The unit shall comply with all MPCB conditions and applicable rules.**

**For Akshay Sansaadhan Pvt. Ltd.**






Authorized Signatory
Name: Abhijeet Gengane
Designation: Founder & CEO
Date: 01-02-2026
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